

Genetic Algorithm for Image Reconstruction

Using 100 Triangles to Reconstruct *Girl with a Pearl Earring*

Tiago Duarte, Sílvia Ferrão, Anastasiia Shulha, Miguel Venancio

Group C20_Fitness

Nova IMS — Computational Intelligence for Optimization

May 2026

github.com/vehnier/GirlPearlEarring-CIF0

1 Representation

We encoded each individual as a real-valued matrix of shape $(100, 10)$, where each row represents one semi-transparent triangle. The ten genes per row are the (x, y) coordinates of three vertices (six values) and four RGBA colour channels — red, green, blue, and alpha — each ranging continuously in $[0, 255]$. We included the alpha channel so that triangles blend rather than simply occlude one another, giving the algorithm more expressive power without increasing triangle count.

Triangles are rendered in row order onto a black 300×400 canvas using alpha compositing, so earlier rows lie beneath later ones. Drawing order is therefore implicitly part of the genome: the GA can learn to place large background-coloured triangles early and fine-detail triangles late.

Alternatives we considered. We evaluated a discrete colour palette but rejected it because it would require far more triangles to approximate the painting’s smooth gradients. We also explored encoding each triangle as a centre point plus circumradius, but that proved geometrically less expressive and provided no natural crossover semantics. The vertex-plus-RGBA representation lets geometric and chromatic genes be handled uniformly by the same operators, which is why we chose it.

2 Fitness Function

We used pixel-wise Root Mean Square Error (RMSE) between the rendered individual and the target image, computed over all $N = 300 \times 400 \times 3 = 360,000$ values:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (p_i - t_i)^2} \quad (1)$$

where p_i are rendered pixel values and t_i are target pixel values. Lower RMSE indicates a better approximation. A random individual scored 77.73; a blank black canvas scored 108.78,

establishing the scale of the problem.

Computational cost. A single evaluation took approximately 75.6 ms (≈ 13 evaluations per second), making large populations expensive. To keep our 3,000-generation final run tractable, we used a *fast fitness* mode that downscales the canvas during evolution and evaluates at full resolution only at the end. We verified that this introduced no bias in operator rankings across sweep experiments.

3 Genetic Operators

3.1 Selection

We tested three methods in a systematic sweep of 84 runs (multiple seeds, 200 generations each). Results are summarised in Table 1.

Method	Mean Best RMSE	Std Dev
Tournament ($k = 3$)	55.87	1.62
Rank	57.65	3.72
Roulette	66.02	1.75

Table 1: Selection method comparison (mean over 3 seeds, 200 generations).

Roulette selection performed worst: early fitness variance caused premature convergence onto a narrow region of the search space. We used tournament ($k = 3$) as the default during exploration sweeps. For the final run we switched to **rank selection**, which is scale-invariant and showed lower sensitivity to population composition over long horizons — an important property when running for 3,000 generations. We also confirmed that $k = 2$ was noticeably worse than $k = 3$ or $k = 10$, so some selection pressure is necessary.

3.2 Crossover

Table 2 compares the four crossover operators we evaluated.

Method	Mean Best RMSE	Std Dev
Uniform crossover	51.36	1.19
Gene-level uniform	52.55	0.71
Gene-level single-point	53.08	2.13
Single-point	55.87	1.62

Table 2: Crossover operator comparison.

We adopted **uniform crossover** (rate 0.80). Single-point crossover performed worst: cutting the chromosome at an arbitrary position mixes genes from triangles at opposite ends of the individual, destroying any positional structure that may have emerged during evolution. Uniform crossover treats each gene independently, making it position-agnostic — appropriate here because no privileged gene position exists in our representation.

3.3 Mutation

Mutation was the single most impactful design choice throughout our experiments (Table 3).

Method	Mean Best RMSE	Std Dev
Vertex jitter	45.38	1.90
Adaptive Gaussian	54.51	1.80
Gaussian ($\sigma = 10$)	55.87	1.62
Colour perturbation	59.68	3.44

Table 3: Mutation operator comparison (all other parameters fixed).

Vertex jitter — perturbing only the three vertex coordinates while leaving colour genes unchanged — outperformed all alternatives by a large margin. This tells us that triangle placement matters far more than colour tuning: once a triangle is near the right position, colour can be refined through recombination with other individuals; but a badly-placed triangle wastes area regardless of colour.

We also swept the mutation rate over $\{0.01, 0.05, 0.10, 0.30\}$. Lower rates consistently outperformed higher ones (mean RMSE 43.79 at rate 0.01 versus 55.87 at rate 0.10), confirming that fine local exploitation is more valuable than broad perturbation at these generation counts.

Multi-mutation strategy (final run). For the final configuration we combined all four operators — vertex jitter, Gaussian, adaptive Gaussian, and colour perturbation — by randomly selecting one per mutation event. This ensemble maintains the dominance of vertex jitter while allowing occasional colour-only or pure-noise mutations that vertex jitter alone would never produce.

3.4 Elitism

We verified that without elitism the best fitness regressed in 21 of 50 generations. We tested elite sizes $\{0, 1, 3, 5\}$ and adopted **elite size 2** for the final run: it guaranteed monotone improvement without over-populating the next generation with near-identical individuals, which would have suppressed diversity.

4 Challenge #3 — Component Contribution Analysis (Ablation Study)

To quantify each component’s individual contribution, we ran five controlled variants over five independent seeds (42, 123, 456, 789, 1337), keeping all other parameters fixed. Results are shown in Table 4.

Mutation is the dominant operator (+16.78 RMSE without it). It is the sole mechanism for introducing new geometric configurations; without it the population converges to a local optimum and barely improves after generation ~ 100 . This is consistent with the continuous, high-dimensional nature of our search space, where crossover alone cannot escape basins of attraction.

Variant	Selection	Crossover	Mutation	Mean RMSE	Std Dev
Full GA	Normal	✓	✓	23.13	0.47
No Crossover	Normal	×	✓	25.54	0.63
Mutation Only	Random	×	✓	29.56	0.41
Random Selection	Random	✓	✓	35.46	1.21
No Mutation	Normal	✓	×	39.91	0.99

Table 4: Ablation study results (mean and standard deviation over 5 seeds).

Selection pressure ranks second (+12.33 RMSE without it). Random selection provides no guidance, so the population drifts. Notably, the *Mutation Only* variant (random selection, no crossover, RMSE 29.56) outperforms *Random Selection* (random selection, with crossover, RMSE 35.46) — showing that crossover is actively harmful without selection, as it recombines poor solutions as readily as good ones.

Crossover is beneficial but secondary (+2.41 RMSE without it). Its value lies in assembling partial solutions from different individuals. The Full GA also has the *lowest variance* across seeds (std 0.47 versus 0.99 for No Mutation), demonstrating that combining all three components yields not only the best mean result but also the most reliable convergence. These rankings were perfectly consistent across all five seeds.

5 Final Configuration and Results

5.1 Configuration

Drawing on every finding above, our final configuration is presented in Table 5.

Parameter	Value	Rationale
Population size	150	Larger populations reduced variance; time is the binding constraint.
Generations	3,000	Convergence curve still declining at generation 3,000.
Selection	Rank	Scale-invariant; stable over long runs.
Crossover	Uniform, rate 0.80	Best performer in sweep; position-agnostic.
Mutation	Multi (vertex + Gaussian + adaptive + colour), rate 0.05	Ensemble keeps vertex jitter dominant while allowing colour refinement.
Elite size	2	Guarantees monotone improvement without crowding.
Local search	5 hill-climbing steps on best individual per generation	Fine-tunes the elite without extra evaluations elsewhere.
Fitness mode	Downscaled during evolution; full-resolution at end	Reduces per-generation cost $\approx 4\times$ with negligible bias.

Table 5: Final GA configuration and rationale for each choice.

5.2 Convergence

Table 6 shows the best RMSE at key checkpoints during the final run.

Generation	100	300	500	1,000	1,500	2,000	2,500	3,000
Best RMSE	38.40	30.93	29.52	27.99	27.07	26.30	25.95	25.57

Table 6: Best RMSE at selected generations during the final run.

The final full-resolution RMSE was **26.76**, reached in **20.5 minutes**. Starting from a random individual at RMSE 77.73, we achieved a **65.6 %** reduction in error. The convergence curve showed no plateau at generation 3,000, indicating that additional compute would yield further gains.

Limitations. With only 100 triangles and no explicit drawing-order strategy, fine details such as the pearl’s specular highlight and delicate fabric texture are poorly approximated. A dedicated initialisation that places large background-coloured triangles first would likely accelerate early convergence, and increasing triangle count would improve quality at the cost of a larger search space.

6 Conclusions

Our systematic sweep and ablation study produced a clear, reproducible ranking of design choices. In decreasing order of impact: (1) mutation method — vertex jitter dominates; (2) mutation rate — lower rates favour exploitation; (3) selection mechanism — rank or high-pressure tournament; (4) crossover method — uniform; (5) elite size — small but non-zero.

The ablation study (Section 4) provided the most direct evidence: removing mutation degrades the GA to near-random search; removing selection makes crossover counterproductive; removing crossover costs only 2.41 RMSE points but increases variance across seeds. The full GA combining all three components achieves the best mean *and* lowest variance, confirming that each component plays a complementary role even when its individual contribution appears modest.